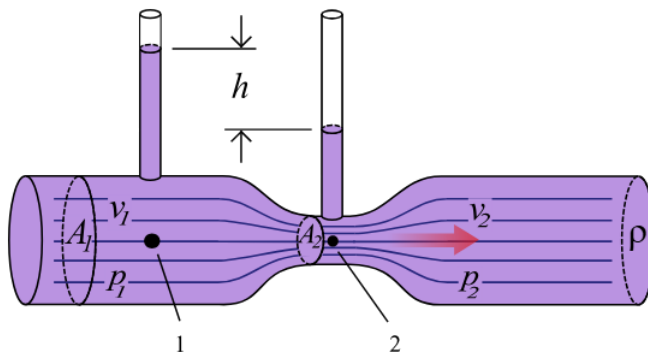


Venturi effect + aerodynamic drag -

<https://www.youtube.com/watch?v=GMmNKUIXXDs>



CoM and CoE explain a venturi - in order to conserve mass, when the pipe narrows the flow must speed up. Then, to conserve energy, in the faster flowing region, the pressure must drop (Bernoulli's). This explains boundary layers around objects in flowing fluid. With a fairly blunt object like a sphere, the fluid flowing past it speeds up as it approaches the top/bottom of the sphere since it is being squeezed past

the free stream fluid and the sphere's surface. The pressure gradient here is favorable (decreasing). Then, as the fluid passes towards the back of the sphere, it slows down and the pressure increases. At some point the increasing pressure will force some of the slow moving fluid next to the surface to reverse direction and separate from the sphere's surface. This creates a wake and a low pressure area behind the sphere. When the separation occurs depends on if the boundary layer was laminar or turbulent. See below.

Laminar vs turbulent boundary layers -

https://www-mdp.eng.cam.ac.uk/web/library/enginfo/aerothermal_dvd_only/aero/fprops/introvisc/node8.html

Turbulent flow mixes more efficiently than laminar which results in the exchange of mass, momentum, etc. This is why a turbulent boundary layer can resist separating more than a laminar boundary layer, since momentum will be mixed which lets the air closest to the surface resist the adverse pressure gradient longer. See golf balls and dimples.

Skin (viscous) drag vs form (pressure) drag -

<https://www.comsol.com/blogs/why-do-golf-balls-have-dimples>

Dimples on a golf ball are helpful because the form drag >>>> skin drag, dimples disrupt the boundary layer and make it turbulent which reduces form drag and therefore overall drag

For application to VAWT blades, this would only be helpful if the form drag is more of a problem than skin drag. But the blades are usually tapered so they're more aerodynamic which reduces the form drag so dimples/texture might not be useful.

Static and dynamic stall -

<https://www.flyaeroguard.com/learning-center/airfoil-stalls/>

If the angle of attack is too large the boundary layer separates much closer to the leading edge of the airfoil which dramatically reduces lift and increases drag. As the blades of an H type darrieus VAWT rotate, the AOA changes, so with fixed blades there is only one point of their path with an ideal AOA and depending on the TSR there may or may not be a point where dynamic stall occurs. This causes a lot of mechanical stress in addition to the loss of lift.

VAWT aerodynamics

<https://www.youtube.com/watch?v=ldt405jIR0E>